

8.2 Classwork: Rational Functions — Toward the Maturità — Markscheme

1. Consider $f(x) = \frac{x+3}{x-2}$.

(a) Denominator $x - 2 = 0 \Rightarrow x = 2$. **Vertical asymptote:** $x = 2$.

(b) Long division: $x + 3 = (x - 2) + 5$, so

$$f(x) = \frac{x-2}{x-2} + \frac{5}{x-2} = 1 + \frac{5}{x-2}.$$

As $|x| \rightarrow \infty$, $5/(x-2) \rightarrow 0$, so $f(x) \rightarrow 1$. **Horizontal asymptote:** $y = 1$.

(c) x -intercept: $f(x) = 0 \Rightarrow x + 3 = 0 \Rightarrow x = -3$. Point $(-3, 0)$.

y -intercept: $f(0) = \frac{3}{-2} = -\frac{3}{2}$. Point $(0, -\frac{3}{2})$.

(d) Right branch ($x > 2$): from $+\infty$ at $x = 2^+$ down to $y = 1$ from above. Through $(3, 6)$, $(7, 2)$.

Left branch ($x < 2$): from $y = 1$ from below at $x = -\infty$ down to $-\infty$ at $x = 2^-$. Through $(-3, 0)$, $(0, -3/2)$, $(1, -4)$.

2. $f_a(x) = \frac{x-2}{x-a}$, $a \neq 2$.

(a) Denominator $= 0 \Rightarrow x = a$. **VA:** $x = a$.

(b) Leading-term comparison: $\lim_{x \rightarrow \pm\infty} \frac{x-2}{x-a} = 1$. **HA:** $y = 1$, **independent of a .**

(c) $f_a(2) = \frac{0}{2-a} = 0$ (for $a \neq 2$). Every curve in the family passes through the point $(2, 0)$.

(d) With $a = -1$: $f_{-1}(x) = \frac{x-2}{x+1}$. VA $x = -1$; HA $y = 1$; y -intercept $f_{-1}(0) = -2/1 = -2$. x -intercept at $x = 2$ (the shared point from (c)).

Left branch ($x < -1$): from 1^+ at $-\infty$ up to $+\infty$ at -1^- ; passes through $(-2, 4)$.

Right branch ($x > -1$): from $-\infty$ at -1^+ up through $(0, -2)$, $(2, 0)$, approaching $y = 1$ from below.

3. $g(x) = \frac{x^2 + x}{x^2 + 1}$.

(a) $x^2 + 1 \geq 1 > 0$ for every real x ; the denominator is never zero. **Domain:** $\mathbb{R}$.

(b) Dividing numerator and denominator by x^2 :

$$g(x) = \frac{1 + 1/x}{1 + 1/x^2} \rightarrow 1 \quad \text{as } x \rightarrow \pm\infty.$$

HA: $y = 1$.

(c) x -intercepts: $x^2 + x = x(x + 1) = 0 \Rightarrow x = 0$ or $x = -1$. Points $(-1, 0)$ and $(0, 0)$.

y -intercept: $g(0) = 0$ (same point).

(d) Table of values:

x	-3	-2	-1	0	1	2	3	5
$g(x)$	$\frac{6}{10} = 0.6$	$\frac{2}{5} = 0.4$	0	0	1	$\frac{6}{5} = 1.2$	$\frac{12}{10} = 1.2$	$\frac{30}{26} \approx 1.15$

Sketch: smooth curve, no asymptote on the x -axis side. Small dip below the x -axis between $x = -1$ and $x = 0$ (local minimum ≈ -0.21 at $x = 1 - \sqrt{2} \approx -0.41$); local maximum ≈ 1.21 at $x = 1 + \sqrt{2} \approx 2.41$; approaches $y = 1$ from both sides.

4. $f_a(x) = \frac{x^2 - ax}{x^2 - a}$, $a \neq 0$.

(a) Domain: $x^2 - a \neq 0$.

(i) $a < 0$: $x^2 - a = x^2 + |a| > 0$ always. **Domain** $\mathbb{R}$.

(ii) $a > 0$: $x^2 - a = 0$ at $x = \pm\sqrt{a}$. **Domain** $\mathbb{R} \setminus \{\pm\sqrt{a}\}$, with vertical asymptotes there.

(b) Leading terms $x^2/x^2 = 1$. **HA:** $y = 1$.

(c) Solve $f_a(x) = 1$:

$$\frac{x^2 - ax}{x^2 - a} = 1 \Rightarrow x^2 - ax = x^2 - a \Rightarrow -ax = -a \Rightarrow x = 1$$

(valid for $a \neq 0$; we need $a \neq 1$ to avoid the removable cancellation at $x = 1$). Then $f_a(1) = (1 - a)/(1 - a) = 1$. **Common point:** $(1, 1)$.

(d) $f_a(0) = \frac{0}{-a} = 0$ for every $a \neq 0$.

Quotient rule:

$$f'_a(x) = \frac{(2x - a)(x^2 - a) - (x^2 - ax)(2x)}{(x^2 - a)^2}.$$

At $x = 0$: numerator $= (-a)(-a) - 0 = a^2$; denominator $= (-a)^2 = a^2$. Therefore $f'_a(0) = 1$ for every $a \neq 0$.

Tangent at the origin: $y = x$, same for every a .

5. $h(x) = \frac{x^3 + 4}{x^2}$.

(a) Denominator vanishes at $x = 0$. **Domain** $x \neq 0$; **vertical asymptote** $x = 0$.

(b) $\frac{x^3 + 4}{x^2} = \frac{x^3}{x^2} + \frac{4}{x^2} = x + \frac{4}{x^2}$.

(c) As $|x| \rightarrow \infty$, $4/x^2 \rightarrow 0$, so $h(x) - x \rightarrow 0$, meaning the graph approaches the line $y = x$.

(d) Table of values:

x	-3	-2	-1	-0.5	0.5	1	2	4
$h(x)$	$-\frac{23}{9} \approx -2.56$	-1	3	15.5	16.5	5	3	4.25

(e) Right branch ($x > 0$): from $+\infty$ at 0^+ down to local minimum $(2, 3)$, then up toward $y = x$ from above.

Left branch ($x < 0$): monotonically increasing from $-\infty$ along $y = x$ (from above), through $(-\sqrt[3]{4}, 0) \approx (-1.59, 0)$, up to $+\infty$ at 0^- .

6. PROBLEMA 2 — Sessione ordinaria 2023.

(a) Domain by sign of a :

$a < 0$: $x^2 - a > 0$ always; domain $\mathbb{R}$, no vertical asymptote.

$a > 0$, $a \neq 1$: domain $\mathbb{R} \setminus \{\pm\sqrt{a}\}$, vertical asymptotes $x = \pm\sqrt{a}$.

$a = 1$: both numerator and denominator factor: numerator $= x(x - 1)$, denominator $= (x - 1)(x + 1)$. **Removable discontinuity** at $x = 1$; vertical asymptote at $x = -1$.

Horizontal asymptote: $\lim_{x \rightarrow \pm\infty} \frac{x^2 - ax}{x^2 - a} = 1$, so $y = 1$ **for every** a .

- (b) Intersection with HA $y = 1$: $f_a(x) = 1 \Rightarrow x^2 - ax = x^2 - a \Rightarrow x = 1$. Then $f_a(1) = (1 - a)/(1 - a) = 1$ (valid for $a \neq 1$). Every Ω_a (with $a \neq 1$) meets the line $y = 1$ at the common point $(1, 1)$. ■

Tangent at the origin: $f_a(0) = 0$ and, by the quotient-rule computation done in Problem 4(d), $f'_a(0) = 1$ for every $a \neq 0$. The tangent line at the origin is therefore $y = x$ for every a . ■

7. PROBLEMA 1 — Sessione ordinaria 2024.

(a) $f_{a,b}(x) = \frac{ax^3 + b}{x^2} = ax + \frac{b}{x^2}$, so $f'_{a,b}(x) = a - \frac{2b}{x^3}$.

Line t : $y = -7x + 12$, slope -7 , value at $x = 1$ is 5 . So $P = (1, 5)$.

Conditions for tangency at P :

- P on the graph: $f_{a,b}(1) = a + b = 5$.
- Slopes match: $f'_{a,b}(1) = a - 2b = -7$.

Subtracting: $3b = 12 \Rightarrow b = 4$; then $a = 1$.

$a = 1$, $b = 4$.

(b) Study of $f(x) = \frac{x^3 + 4}{x^2} = x + \frac{4}{x^2}$.

- Domain $x \neq 0$; vertical asymptote $x = 0$.
- Oblique asymptote $y = x$ (from $f(x) - x = 4/x^2 \rightarrow 0$); curve sits above the asymptote on both sides since $4/x^2 > 0$.
- $f'(x) = 1 - 8/x^3 = 0 \Rightarrow x = 2$. Local minimum $(2, 3)$. On the right branch, f decreases on $(0, 2)$ and increases on $(2, \infty)$. On the left branch, $f'(x) = 1 - 8/x^3 > 1 > 0$ for $x < 0$, so f is monotonically increasing on $(-\infty, 0)$.
- $f''(x) = 24/x^4 > 0$ for $x \neq 0$; curve is convex on each branch.
- Zero of f : $x^3 + 4 = 0 \Rightarrow x = -\sqrt[3]{4} \approx -1.59$. The only x -intercept.
- As $x \rightarrow 0^\pm$, $4/x^2 \rightarrow +\infty$, so $f \rightarrow +\infty$ from both sides.
- As $x \rightarrow +\infty$: $f \rightarrow +\infty$ along $y = x$. As $x \rightarrow -\infty$: $f \rightarrow -\infty$ along $y = x$.

Sketch summary: two branches, both convex, both asymptotic to $y = x$ as $|x| \rightarrow \infty$ and to $x = 0$ as $x \rightarrow 0$. Right branch has a minimum at $(2, 3)$ and lies in the first quadrant; left branch monotonically rises from $(-\infty, -\infty)$ through $(-\sqrt[3]{4}, 0)$ to $+\infty$ at 0^- .